



Benefits realisation in an agile environment

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ABSTRACT

Success frameworks from an information systems and project perspective include benefits realisation as a measure of success. Research highlights benefits are not realised resulting in low IT project success rates. Benefits realisation intrinsically forms part of agile as benefits are delivered frequently to customers resulting in higher success rates. What is not known, is to what extent organisations, that are scaling agile, are implementing benefits realisation management to improve the success rates of their IT projects. Semi-structured interviews were conducted with nine experts from organisations that adopted a scaled agile framework. The results show that benefits realisation is performed unsystematically, and the success of IT projects is attributed to the agile approach and not to BRM. Twelve principles were identified to facilitate agile BRM and ultimately improve project success. This article provides new insights into the BRM debate through its novel approach of linking agile, BRM and project success.

1. Introduction

A positive relationship between benefits realisation management (BRM) and project success has been established by various authors (Ul Musawir et al., 2017; Badewi, 2016; Gomes & Romão, 2016). Benefits realisation management (BRM) ensures that benefits associated with information technology (IT) projects are achieved and sustained. However, the practical application of BRM is not achieving what it set out to achieve. Organisations do not necessarily harvest the intended benefits of an IT project as there is a break in linking organisational strategies, the rationale of the project and the intended benefits (Marnewick, 2016; Davis, 2017). The BRM process is well documented and there are various models and frameworks. BRM comes with its own set of problems ranging from non-adherence to the BRM processes (Marnewick, 2017), confusion around the responsibility of benefits realisation (Zwikael et al., 2019), the formulation and appraisal of target benefits (Zwikael et al., 2018) as well as the intersection of the BRM process with the traditional project life cycle.

Research further indicates that the adoption of benefits management and agile practices has a positive impact on realising benefits due to agile practices' frequent deployment and flexible scope (Holgeid & Jørgensen, 2020). Some difficulties in adopting BRM in agile projects

have been highlighted, such as the requirements for a business case (Terlizzi et al., 2017), and long-term benefits should be validated which implies a closer resemblance to traditional project life cycles (Lappi et al., 2018).

Project success has evolved from measuring the triple constraint to a more complex multi-dimensional measurement. Various complex and multi-dimensional models and frameworks are available to measure the success of a project. A two-dimensional perspective investigates project management and product success as the success criteria (Camilleri, 2011; Ika, 2009). Project management success focuses on the project management process itself and has three success criteria, i.e. (i) time, cost and quality, (ii) the quality of the project management process and (iii) stakeholder satisfaction. Product success focuses on (i) realising the project's goal, (ii) realising the purpose of the product and (iii) stakeholder satisfaction in terms of the final product. Shenhar (2001) brought a four-dimensional framework to the table focusing on project efficiency, the impact on the customer, business success and preparing the organisation for the future. Bannerman and Thorogood (2012) created a five-level framework to determine project success specifically focusing on IT projects. The five levels are (i) process success, (ii) project management success, (iii) product success, (iv) business success and (v) strategic success. These different interpretations of measuring success

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are a concern to the various stakeholders. Each stakeholder has its own perception of whether a project is successful or not, as well as how success should be measured (Davis, 2017). There are overlaps between the different models and frameworks on how to measure project success, but the focus has shifted towards the benefits that organisations harvest from their investments in projects.

Apart from the framework of Bannerman and Thorogood (2012) that focuses specifically on measuring IT success from a project management perspective, project management literature focuses on IT project success criteria and critical success factors. A causal model was developed by Petter et al. (2008) to specifically measure the success of an information system. As with the framework of Bannerman and Thorogood (2012), 'net benefits' was identified as a dependent dimension of Information Systems (IS) success. This implies that benefits realization contributes to IT project success. However, these frameworks and models were derived using traditional project management methodologies such as Waterfall or the V-model. *This creates a gap in the current knowledge as it is not known how benefits are managed and realized in an agile environment and when benefits are not realized, then an IT project cannot be perceived as successful. The focus of this article is on benefits realization as a dependent dimension of IT project success within a scaled agile environment.*

IT projects are notorious for not delivering value and this has created an opportunity for the introduction of agile to improve IT project success. Agile projects do not conform to the various project success models and frameworks (Petter et al., 2008; Bannerman & Thorogood, 2012), as the delivery of the product is a continuous process starting with the minimum viable product (MVP) and further enhancements. The focus is on value delivery and the realisation of benefits. To institutionalise this focus on benefits, a benefits management process is required to facilitate a shared stakeholders' view of the success criteria and their measurement and monitoring (Davis, 2017). Due to the success of agile, organisations are scaling agile. Scaling agile implies that the agile principles, practices and outcomes are applied at all organisational levels.

IT projects using a Waterfall approach, deliver a once-off product and the success of this product is measured based on the process, project management and product criteria. Benefits realisation only starts once the product is operationalised. In contrast to the Waterfall approach, a product is continuously delivered within an agile environment. This is contributed to the iterative nature of agile where the MVP and subsequent enhancements are delivered frequently and iteratively. This implies that IT project success and IS success is measured much earlier and more frequently. Benefits realisation also occurs earlier in the lifecycle i.e. as soon as the MVP is delivered. Benefits realisation is then an iterative process as the product itself is changed and enhanced. This is in contradiction with the BRM process associated with the Waterfall approach. This article addresses this anomaly and focuses on the following research problem: *does BRM implemented in a scaled agile environment ensures IT project success?* The following sub-questions guides the investigation:

- 1 What is the relationship between BRM and project success?
- 2 How does the BRM process differ when an agile approach is used instead of a Waterfall approach?

This article consists of six sections. The 'Introduction' section presents the background. The section 'Literature review' deals with what BRM is, as well as challenges currently experienced. The framework derived from the literature review is also depicted. The 'Research methodology' section deals with the research methodology, research design, tools used to collect the data and analysis method. The section 'Results and analysis' presents the findings and in the 'Discussions' section the results from the research are linked to the literature review. Lastly, conclusions drawn from the study are presented.

2. Literature review

Measuring the success of a project has evolved over decades from the triple constraint to delivering strategic intent and realising benefits. Ika (2009) divided the evolution into three periods, ranging from the 1960s to the beginning of the 21st century, and concludes that there has been a shift from measuring projects only according to time, cost and quality to a more holistic approach focusing on benefits to the stakeholders. Bannerman and Thorogood (2012) developed success measures specifically for IT projects that span five levels that is in line with the findings of Ika (2009) as well as the four dimensions of Shenhar et al. (2001). The project success measures are mapped against the five levels as indicated in Table 1.

To ensure success at all five levels, Badewi (2016) suggests that project management and BRM be integrated. Project management ensures success at the process, project management and product levels, whereas BRM ensures success at the business and strategic levels. The focus of IT projects should be on realising benefits rather than the actual artefact (Doherty et al., 2012). Projects create a product or service (Project Management Institute, 2021) which is then utilised within the organisation to bring about change (Zwikaal et al., 2019). This change should be positive and is consequently referred to as the benefit of the product. Benefits can be perceived as either positive or negative (dis-benefit). They arise primarily from organisational change, including improved information usage, that accompanies an IT implementation, rather than directly from the technology itself (Doherty et al., 2012).

DeLone and McLean (2003) developed a framework to specifically measure the success of an information system. This framework was enhanced (Petter et al., 2008) where the focus of the sixth dimension is on 'net' benefits. 'Net' benefits are perceived as "the extent to which IS are contributing to the success of individuals, groups, organizations, industries, and nations" (Petter et al., 2008). Examples of benefits are improved decision-making, the increase of sales or improved profits. The 'Net' benefits dimension contributes to the success of an information system i.e. the more benefits, the more successful the information system. The IT project success levels of Bannerman and Thorogood (2012) include the realisation of benefits as part of business success i.e. the more benefits that are realized at the business level, the more successful the IT project itself. Fig. 1 provides a mapping of the information systems success dimensions to the IT project success levels. Four of the dimensions can be mapped to the product success level and the 'net' benefits level is mapped to the business success level.

The logical conclusion is that benefits realisation forms part of IT project success and that the product (information system) should provide benefits to the organisation at large. Therefore, benefits realisation is a measurement of IT project success both from an information systems perspective as well as a project perspective.

2.1. Rationale for benefits realisation

The argument is that the intended benefits that are promised within the business case (Einhorn et al., 2019) are realised once the project's deliverable brings about the anticipated change. The process of managing benefits from the point that they were identified in the business case to the point where they are perceived as actual benefits is called benefits realisation management (BRM). BRM has been researched by various authors. Holgeid and Jørgensen (2020) did a survey of 71 Norwegian software professionals' adoption of benefits management and agile practices. Based on this analysis, they developed a five-step BRM process. Cha et al. (2018) created a six-step project management knowledge framework for organisational transformation. The purpose of this framework is to highlight how IT projects can provide benefits to the organisation. The adoption of BRM in Brazilian financial institutions was investigated by Terlizzi et al. (2017) and they identified six practices and seven barriers affecting the adoption of IT BRM. Fernandes and O'Sullivan (2021) investigated BRM from the perspective of a large

Table 1
Success levels and associated measures.

	Levels	Measures	Authors
Before the project	Benefits identification and motivation for the project	• Business case • Identified benefits • Benefits realisation plan	Badewi (2016) Holgeid et al. (2021) Einhorn et al. (2022) Zwikael et al. (2018)
During the project (Focus on the product)	Process success focuses on the technical and managerial processes required during a project's life cycle. Examples are configuration management, quality control and risk management. Project management success measures time, cost, scope and quality. Product success relates to the customer since this measurement focuses specifically on ensuring that the project meets the needs of the various stakeholders. Empirical indicators include specifications met, client/user expectations met, and product/system used.	• Fit for the purpose • Aligned with project objectives • Integrated (as appropriate) • Effectively implemented • Schedule met • Budget not exceeded • Project scope achieved • Specifications met • Requirements met • Client/user expectations met • Client/user acceptance • Product/system used • Client/user satisfied • Client/user benefits realized • Meeting functional performance • Fulfilling customer needs • Solving a customer's problem	McLeod et al. (2012) Marnewick et al. (2017) Savolainen et al. (2012) Kazmi et al. (2016) Jugdev and Müller (2005) Ika (2009) Todorović et al. (2015) Shenhar et al. (1997) Shenhar et al. (2001) Pinto and Slevin (1987) (Joseph and Marnewick (2014) Cooke-Davies (2002) (Camilleri (2011) Jugdev and Müller (2005) (Marnewick (2012) Marnewick (2013) Joseph et al. (2014) Baccarini (1999) Wateridge (1998) Serrador and Turner (2015) Bennington and Baccarini (2004) Lappe and Spang (2014) Chih and Zwikael (2015) Marnewick (2016) Serra and Kunc (2015) Chih and Zwikael (2015) (Serra and Kunc (2015) Shenhar et al. (2001) (Aubry and Hobbs (2011) Mir and Pinnington (2014) Peterson (2002) Longman and
After the project (Focus on the outcomes of the product)	Business success focuses on how projects are used to realise benefits in relation to organisational objectives. Indicators include the validation of the business case and the realization of benefits. Projects should align with an organisation's strategy to ensure that strategic success is realised. Increase in market share is an indicator.	• Commercial success • Creating a large market share • Objectives met • Business case validated • Business benefits realized • Creating a new market • Creating a new product line • Developing a new technology • Business development enabled • External stakeholder/competitor recognition	Longman and

Table 1 (continued)

Levels	Measures	Authors
	• Competitive response generated	Mullins (2004) Young and Grant (2015)

university-industry collaboration programme. They discussed various models and identified common BRM practices that can be used in university-industry collaborations. Other authors (Badewi, 2016; Zwikael et al., 2018; Lin & Pervan, 2003) derived their own version of BRM based on the original works of Ward and Daniel (2012), Bradley (2010) and Ashurst (2011). All the models, frameworks and processes can be summarised into the following steps: (i) benefits identification, (ii) benefits analysis and planning, (iii) benefits management, (iv) benefits transition and (v) benefits sustainment. The relationship between BRM and the project life cycle is shown in Fig. 2.

Benefits are harvested once the project is completed and the deliverable brings about the intended change. This relates to the fourth and fifth levels as per Table 1. Svejvig and Schlichter (2020) highlight that the linear process typically followed in BRM models is probably not best suited for the complexity associated with benefits realisation. They postulate that this might be the reason for the challenges associated with BRM.

2.2. From agile to scaled agile

IT projects are failing at an alarming rate and organisations do not necessarily harvest any benefits from these projects. Studies such as the Chaos Chronicles (The Standish Group, 2013; The Standish Group, 2018) and the South African Prosperus reports (Labuschagne & Marnewick, 2009; Marnewick, 2013; Marnewick et al., 2017) indicate that a mere 30% - 40% of IT projects are successful.

Due to these poor success rates, an alternative approach was recommended i.e. agile. Agile methodologies were initially formulated to manage the development of software development projects but have been expanded due to their popularity and success to deal with most types of IT projects. Results highlighted that projects embracing agile principles are more successful (Khoza & Marnewick, 2020; The Standish Group, 2018). Agile makes use of iterative processes that are a discovery-based approach. Several adaptive (agile) methods are used during software development such as Adaptive Software Development, Extreme Programming, Scrum, Dynamic Systems Development Method and Crystal Methods (Leffingwell, 2011; Sillitti & Succi, 2005). A high-level view of these approaches is that the code is written very quickly without detailed design, after which it is evaluated by customers and adapted after feedback. This implies that benefits can be realised much sooner and customers do not have to wait until the entire product is delivered. Due to the success of agile, organisations are scaling agile. The reason for scaling agile is to harvest the benefits associated with agile (Dikert et al., 2016). These benefits can be summarised as team morale improvements, increased reports of project predictability and a reduction in project risk (VersionOne Inc., 2019). There are various scaling agile frameworks. Table 2 summarises the most popular and influential scaled agile frameworks.

There are considerable similarities between the different frameworks described in Table 2. In terms of principles, practices and artefacts, they all rely on agile values to deliver solutions (Theobald et al., 2019). However, they present different challenges and have different levels of adoption. The most adopted frameworks are the Scaled Agile Framework (SAFe) and Scrum@Scale (VersionOne Inc., 2019; Khoza & Marnewick, 2021).

Benefits realisation within a scaled agile environment is complex and not necessarily explicitly addressed within the various frameworks. Within SAFe, the concept of value streams is introduced which perform the same function as benefits realisation. Value streams represent the

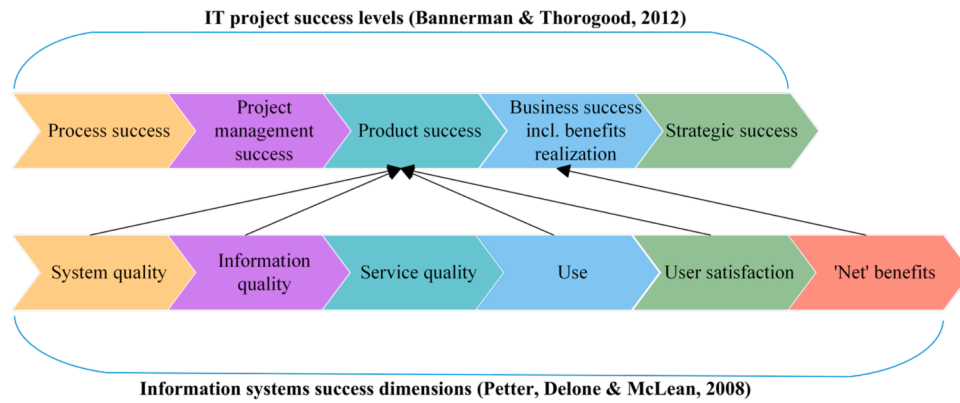


Fig. 1. Mapping of IT project and IS success dimensions.

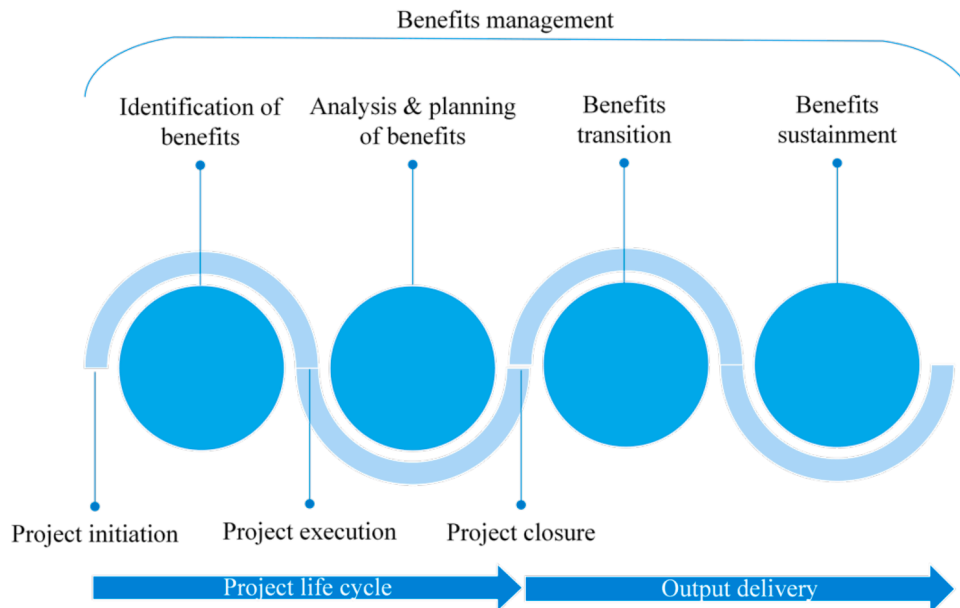


Fig. 2. Traditional benefits realisation process.

processes that an organisation uses to implement solutions that provide continuous value to its customers (Scaled Agile Inc., 2020). A portfolio contains various value streams and each one of these value streams is dedicated to build and support a set of solutions. Identifying value streams and agile release trains (ARTs) includes the following steps: (i) identify the operational value streams, (ii) identify the systems that support the operational value stream, (iii) identify the people who develop and maintain those systems, (iv) define the development value streams that contain the systems and people, (v) add the people needed to build the full business solution and (vi) identify the ARTs that realise the development value streams. It must be noted that each value stream represents a substantial investment. Therefore, governance needs to be in place to evaluate how a value stream is performing against its intended benefits. To that end, each value stream does have a set of key performance indicators (KPIs) that can be used to evaluate the ongoing investment in a particular value stream. The purpose of Scrum@Scale is to “help an organization to focus multiple networks of Scrum teams on prioritized goals” (Sutherland, 2020). The prioritisation of a backlog forms part of the product owner cycle. The purpose of a prioritised backlog is to optimise value delivery. However, Scrum@Scale does not mention how value should be created.

2.3. Value streams

Value streams focus on the process and eliminating waste. The focus is not necessarily on delivering benefits but benefits might be an unintended consequence. To create continuous flow of value, waste should be reduced – the less waste, the more value. Value stream mapping (VSM) is used to identify waste through visualisation and analysis of the current value stream (Ali et al., 2015; Khurum et al., 2014). Waste can be defined as any activity or process that consumes resources, time or space but does not add any value to the product as perceived by the customer (Khurum et al., 2014). At the heart of VSM are cross-functional teams that include members from each level within the organisational structure (Manos, 2006). To create VSM, a dedicated team is required that asks difficult questions and thinks beyond the current state (Manos, 2006). The mapping must be done on real data (Womack, 2006) and the team needs to understand the problem before they start with the mapping process. The team must keep in mind that the customer is only interested in the value that they harvest from using the product. If a product is not adding value, it is most likely just adding *muda* (the Japanese word for waste) (Manos, 2006). Value is determined by the customer or end-user of the product and it is important to have a clear understanding of what that is (Khurum et al., 2014). This relates to project success levels 4 and 5 as well as the ‘Use’ and ‘User satisfaction’

Table 2
Scaled agile frameworks.

Framework	Purpose	Scope	Target audience
Scaled Agile Framework (SAFe) (Leffingwell, 2011)	Provides a knowledge base of proven and integrated principles, practices, artefacts and competencies for achieving business agility.	Software with 50 to 120 people	Large and traditional lean organisations
Scrum@Scale/Scrum of Scrums (SoS) (Sutherland, 2020)	Coordinates teams in a way that optimises the overall strategy of the organisation, by extending across the organisation the way a single scrum team functions.	Software, hardware and systems; flexible with 5 to 10 teams	All types of organisations
Large-Scale Scrum (Less) (Larman & Vodde, 2016)	A framework that scales scrum principles, purpose and elements to apply in a larger scaled context.	Software with up to 10 teams of 7 members each	Large organisations
Disciplined Agile Delivery (DAD) (Ambler & Lines, 2012)	A learning-oriented agile approach to develop IT solutions that mix multiple agile methods and include extra practices to adapt and scale agile to an enterprise level.	Software with 200 people or more	Multiple organisations
Spotify (Kniberg & Ivarsson, 2012)	Provides a people-driven, autonomous set of agile practices that reflect value creation which scales while stressing the importance of culture and networks.	250 to 300 people	Multinational organisations like Spotify

dimensions of the IS success framework. Within an agile environment, four items influence the flow of value to customers: (i) the feature itself that creates visibility to the customer, (ii) a defect or bug in the system, (iii) risk that includes security, regulatory and compliance work and (iv)

technical debt (Cunningham, 1992). In a traditional environment, these four items are only addressed once the entire product is delivered. Users cannot test features and make changes until the final product is delivered which create rework (technical debt). Users might experience less value from the final product using a traditional approach such as Waterfall.

2.4. A conceptual model for scaled agile benefits realisation

Fig. 3 illustrates the conceptual model that forms the basis of the research and is derived from literature. The first level, *Portfolio*, is concerned with the organisational strategy and the value streams. This is a continuous process where the strategy is divided into value streams. A value stream is the primary construct for understanding, organising and delivering benefits in a scaled agile environment.

The second level, *Planning and Prioritisation*, focuses on the benefits and how they are identified, planned, transitioned and sustained. This is in line with traditional BRM as per Fig. 2. The value case serves as input. The re-evaluation and reprioritisation of projects and their associated benefits is a frequent activity and is normally done at a program increment (PI).

At the *Implementation* level, normal agile implementation occurs using methodologies such as Scrum. The product backlog is used to determine the sprints and the actual delivery of the product. This product should deliver the intended benefits as identified during the *Planning and Prioritisation* level.

It is evident from Figs. 2 and 3 that there are similarities between traditional and agile BRM. Benefits arise primarily from organisational change, including improved information usage that accompanies an IT project and the approach to how this change process impacts benefits realisation (Doherty et al., 2012). This comparison is illustrated in Table 3.

BRM in a traditional setting follows a very linear process where the business case upfront determines the priority of the project. The intended benefits are only realised once the project deliverable institutes change into the organisation. Although the business case should be re-evaluated periodically to determine the validity of the benefits, this is not done (Einhorn et al., 2019; Einhorn et al., 2022). In a scaled agile environment, the value case is continuously re-evaluated at the PI planning sessions and value is continuously added to the organisation

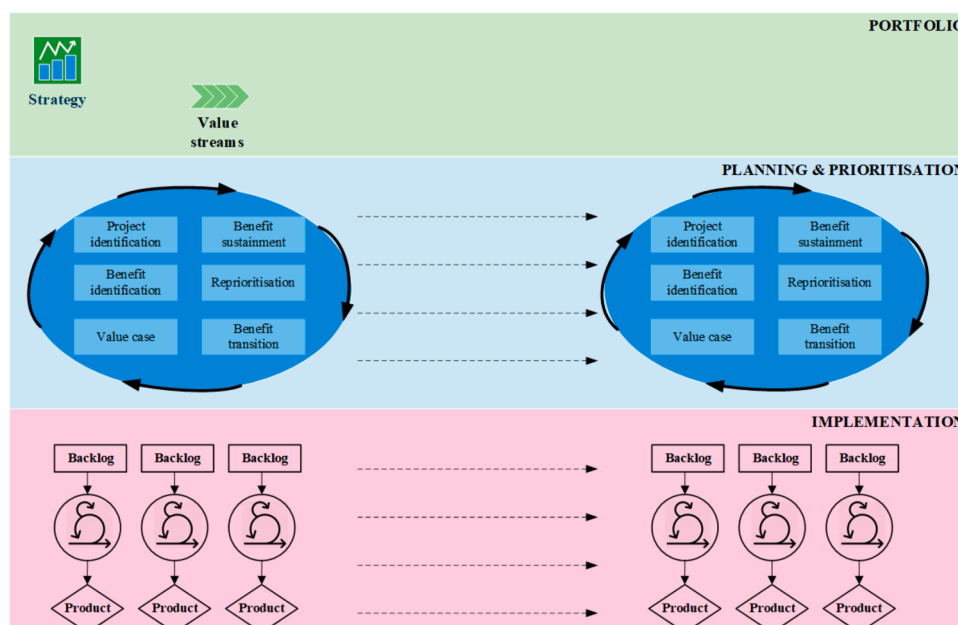


Fig. 3. Agile benefits realisation process.

Table 3

Traditional and agile BRM comparison.

BRM activities (Holgeid & Jørgensen, 2020; Fernandes & O'Sullivan, 2021)	Traditional BRM	Agile BRM
1. <i>Identifying and structuring benefits</i> (Focus on benefit activities and outcomes should be defined in terms of benefits to be realised versus functionality of technology.)	• The business case defines the benefits and how they will be measured (Badewi, 2016).	• Pre-investment appraisal to be done before start of project (Holgeid et al., 2021). • Re-estimation of benefits and identification of benefits after completion (Holgeid & Jørgensen, 2020). • Value streams level introduced by scaling agile (Ebert & Paasivaara, 2017).
2. <i>Planning benefits realisation</i> (Activities need to realise benefits, including potential organisational and process changes.)	• Benefits identification, planning and implementation before, during and after project (Cha et al., 2018). • Prioritisation of IT projects is done as a once-off exercise (Einhorn et al., 2019).	• Benefits identification, planning and implementation are done during a PI planning session of delivery cycles (Petit & Marnewick, 2021b). • Prioritisation of IT projects is done at multiple PI planning sessions based on the intended value which enables replanning (Neve et al., 2017). • This is a continuous process throughout the project life cycle.
3. <i>Execution of benefits realisation plan</i> (Implementation of benefits plan which is integrated with the project management plan. Benefits management goes together with change management.)	• Benefits realisation and operationalisation normally take place after the project life cycle (Cha et al., 2018). • To leverage the business from new IT functionality, a change management process is facilitated in parallel with project processes. Change management is a separate process, although the project deliverable impacts the change (Doherty, 2014; Badewi, 2016).	• Change is managed through continuous prioritisations, iterative planning using project vision and backlogs (Lappi et al., 2018). • Continuous value delivery (Holgeid & Jørgensen, 2020). • The PI planning sessions are used to continuously evaluate the intended value of a project (Petit & Marnewick, 2021b).
4. <i>Sustainment</i> (Evaluation of benefits delivered and identification of actions to receive missed benefits.)	• Long-term post-project benefits realisation (Lappi et al., 2018).	• Short-term value realisation through the utilisation of deliverable from short delivery cycles (Lappi et al., 2018).

through the incremental change associated with agile and scaled agile. To conclude the literature, the following can be derived:

- Benefits realisation is a measurement of project success (Bannerman & Thorogood, 2012; Petter et al., 2008).
- BRM is not done within traditional IT projects resulting in low success rates (Marnewick, 2016; Marnewick, 2017).
- Agile projects are perceived as more successful than traditional IT projects (Marnewick et al., 2017; The Standish Group, 2018). Yet, Khoza and Marnewick (2020) highlighted that a Waterfall or an agile approach is equally suitable to realise benefits.
- There are no foundational differences between traditional BRM and agile BRM as per Table 3. The difference lies within the fact that agile is iterative and that the BRM process should also be iterative.

- BRM is inherently part of agile due to the Agile Manifesto's principles, iterations and involvement of stakeholders. Little or no evidence exist whether BRM is done in a scaled agile environment and if it is done, how it is done. This is the identified gap that this article is addressing.

The research methodology that was used to answer the research question is elaborated on in the next section.

3. Research methodology

A qualitative approach was used to answer the research questions. This approach seemed more appropriate as the research was of an exploratory nature (Yin, 2017). A semi-structured interview guide was developed based on previous studies on BRM (Marnewick, 2016) as well as constructs from agile and scaled agile. The semi-structured interview consisted of 24 questions divided into four sections. The process of Castillo-Montoya (2016) was followed to align the questions with the research question.

- 1 Section 1 focused on biographical information. As part of introduction each interviewee were asked to provide a brief background on their experience with agile project implementations.
- 2 Section 2 focused on the agile benefits realisation process and relates to the research question addressed in this article.
- 3 Sections 3 and 4 focused on benefits ownership and the challenges associated with agile benefits realisation. This is not part of this article.

Cunningham (2008) and Kwok and Ku (2008) suggest semi-structured interviews as being an excellent way to gather detailed information. Interviewees are given the opportunity to elaborate in a way that is not possible with other methods, but they are able to share information in their own words and from their own perspectives. Virtual interviews were conducted due to COVID-19 restrictions and Microsoft Teams was the preferred collaboration tool. Purposive sampling was used to identify the interviewees. This was based on their participation within an agile project management office (PMO) community of practice as well as their roles within their respective organisations' scaled agile transition. The interviewees were from diverse organisations and effort was made to interview more than one representative from the organisation to get a less biased view. The organisations were selected as they are either still in the process to scale agile or completed the process. In certain instances, the interviewee was responsible for the entire agile transition and had in-depth knowledge about the benefits realisation process e.g. CVO-1.

The recorded video files were transcribed verbatim. This was done using Microsoft Teams' transcription functionality as a first run, after which the original transcription was edited by listening to the recordings. The transcriptions went through an iterative process of open coding which summarised the data into smaller meaningful units called codes and themes (Bryman & Bell, 2017; Saunders et al., 2016). The coding process was facilitated using ATLAS.ti. The following process was used:

- One researcher extracted and coded the data elements to allow for comparisons across the interviews. The second researcher compared the coded data elements and resolved differences through playing back recorded interviews and discussions until consensus was reached.
- Codes were chosen as derived by the researchers or *in vivo* codes. A deductive approach was used to search for specific codes relating to agile BRM. An inductive approach was also followed to identify new codes and themes. A continual refinement process was used to merge similar codes as well as discard those deemed to lack sufficient data to support themselves.

Table 4
Summary of interviews.

Type	Role	Reason for selection	Agile experience (years)	Duration & Pages
Financial institution (1)	Head of PMO (PMO-1)	Involved from the beginning in the initial scaled agile transition.	11	00:53:30 (14)
	Head of Transformation (TRAN-1)	Currently involved with further deployments of scaled agile.	8	00:45:38 (11)
Consulting	Partner (PART-1)	Involved in a consultative capacity with the scaled agile transition of various corporates.	11	01:23:49 (18)
Financial institution (2)	Business IT executive (IT-1)	Involved from the beginning in the initial scaled agile transition.	10	00:57:45 (16)
Entertainment company	Change manager (CHANGE-1)	This company is in the beginning of its scaled agile journey. The consultant was involved with previous journeys whereas the other two interviewees are leading the scaled agile transition.	2 2 10	01:03:29 (17)
	Business transformation (TRAN-2)			
Financial institution (3)	Consultant (CONS-1)			
	Chief value officer (CVO-1)	Involved from the beginning in the scaled agile transition and is currently overseeing the entire value creation process.	7	00:39:18 (11)
IT outsourcing	Agile coach (COACH-1)	Involved with the internal scaled agile transition before becoming an agile coach.	10	00:33:06 (11)

- ATLAS.ti's coding and co-occurrence matrices were used to identify relationships between codes and themes as well as new possible themes within the data (Bryman & Bell, 2017).
- Themes were based on the agile benefits realisation process (Fig. 3). Codes were allocated to the themes. For example, 19 codes relating to challenges were identified and grouped into one theme 'Challenges'. A new theme (Principles) emerged with 15 associated codes.
- A further refinement process was followed to remove codes that could not be accurately linked to existing themes. A total of 175 codes remained.

4. Results and analysis

This section is based on Fig. 3 and starts at the *Portfolio* level.

4.1. Portfolio level

As per Petit and Marnewick (2021b), value streams are important as they link the various IT projects with the organisational strategies. The notion of *value streams* was mentioned by the interviewees. Value streams are typically used for resource allocation. This is in contradiction of the theory where value streams are focusing on delivering value and benefits.

The *Planning and Prioritisation* level focuses on BRM, and the focus of the analysis is to determine the level of conformance.

4.2. Planning and prioritisation level

The *business or value case* was used to motivate the rationale of the IT project (Einhorn et al., 2019). The interviewees were familiar with the concept of the business case. IT-1 claimed that, in line with the Agile Manifesto, they had a "light business case" that provided "a much clearer understanding of what those benefits are and our business CFOs plays a critical role in that." COACH-1 elaborated on this notion, stating that "they will set up the test cases to make sure that what they are developing is really, what the client wants at the end of the day and they will put the same metrics in and measure the success from the onset". Although there was familiarity with a value case, the interviewees implied that it was either not fully utilised or was misused.

- Projects were identified by the executive team and based on their instruction, initiatives were implemented without a business case: "... and we are not really kind of putting business cases further than that around it" [PMO-1].
- "Although everybody wants it because you'll find it remember financial managers have no agile training so they don't care, they want something that shows them the cost and the forecast" [PART-1].

An important activity of BRM is to *identify the benefits*. Based on the analysis of the interviews, it can be concluded that this activity was performed.

- "In order to get priority and needs to have a clear understanding what the benefits associated with them are" [IT-1].
- "So it goes around, you know, defining the use case, the user story, the value drivers and actions, and levers that we're going to take" [TRAN-2].

The identification of benefits has a second purpose, i.e. the *prioritisation* of the project in the current pipeline. Prioritisation could and should change as the associated benefits change.

- "And therefore that epic and its benefits kind of understood and kind of agreed and prioritized based on what that would contribute, and that gets pulled into a program increment and the teams have full visibility" [PMO-1].
- "We prioritizing highest value and least effort and then at the same time when we defining a user story to understand what product to deliver, it's going that user story is a key part of the planning, but it feeds into the value we hope to create" [CONS-1].

Prioritisation and reprioritisation occurred typically at a *PI event* or some similar process.

- "So what we do is that you don't necessarily earn the right to magically move into the following PI. So even though that epic span multiple program increments, you have to recontend for priority for the next program increments" [PMO-1].
- "But something that maybe hasn't wasn't there at the start of the sprint cycle, which maybe is more urgent or whatever, so there's ongoing refinement of priorities." [CONS-1].

CVO-1 highlighted some concerns with the PI sessions: *“I think the PI is maybe too siloed at the moment as opposed to cross functional, if that makes sense, and then to the same extent that retrospective process from a PI point of view isn’t working as well as it should be.”*

Projects and their associated benefits should continuously be evaluated to determine whether it still makes sense to continue with the project. Interviewees indicated that this was typically done through **benefits tracking**.

- *“Measurable benefits that we went to track and then coming back six months after we release that to say OK, did we achieve these things and saying whether we were successful or not, we’re not really that that framework isn’t there”* [PMO-1].
- *“... what they would do is they would track and verify based on the measurement criteria and the benefits map”* [PART-1].
- *“Yeah, so that’s something that I’m not measuring”* [TRAN-1].
- *“Uh, and then looking at the metrics and targets that would be tracked around that use case once it’s delivered, and then some sort of measurement depending on what it is.”* [TRAN-2]

There was no consensus on the level of adherence regarding benefits tracking. In some instances, it was done but what is evident from the interviews is that it is not an easy process. *“So first of all the source for tracking is. Sometimes those sources are not well defined especially when it becomes vertical e.g. revenue uptake”*.

Benefits planning was not highlighted as a separate activity by the interviewees and was incorporated into the PI sessions. Benefits are associated with change that is perceived as positive. To harvest the benefits, an organisation needs to have change management processes in place. No evidence could be found indicating that the organisations in this study planned to harvest and **sustain the benefits**.

4.3. Implementation level

At the third level, *Implementation*, the impression was created that the various projects were derived from the PI sessions: *“So we spend a whole day where teams come and talk through what are the intentions and objectives we’re trying to hope to get out of the next program increment”* [IT-1]. Once the team started with the implementation, they followed an incremental process of developing the product. This was typically done through Scrum as a method. Various sprints were part of the PI and the progress influenced the reprioritisation and continuation of the project. The team itself was not concerned with the realisation of benefits, *“But for the people who in the work on a daily basis, I think they just go and deliver. I think that organisational context and connecting the dots is weak to say at best”* [CVO-1], and those teams that were indeed concerned with benefits realisation were typically mature cross-functional teams: *“So what you’ll find is that the teams are involved in understanding why things was prioritized and therefore the product owners also most of them reporting into business and kind of bringing that information back into the teams that conversation is kind of less scheduled but more kind of interactive”* [IT-1].

5. Discussion

Benefits realisation is a difficult process to master. Challenges range from BRM not being fully implemented (Marnewick, 2016) to the difficulty of setting effective benefit targets (Zwikaël et al., 2018) and confusion about who the owner of the benefits is (Zwikaël et al., 2019). The results of this study emphasise that this is still an ongoing trend.

Within a scaled agile environment, value streams are the vehicles to implement strategies (Petit & Marnewick, 2021b). This is done to optimise the value and to reduce waste. Organisations in this study use value streams for the allocation of resources and this allocation changes as the anticipated value of a stream changes. Resources are then re-allocated to streams that offer more value. The value streams are also used for budgeting purposes and forecasting. A benefit that the interviewees

agreed on is the visibility associated with value streams. Everyone in the organisation can see the strategies, their associated benefits and the resources allocated to them. This in turn creates unintended governance decisions and the consequences are transparent. The results correspond with theory highlighting that value streams reduce waste through visualisation which in turn creates transparency (Ali et al., 2015; Khurum et al., 2014).

Literature suggests that a value case informs the benefits associated with a project (Einhorn et al., 2019; Einhorn et al., 2022). Different perspectives are captured in the value case ranging from a financial view to the actual value of the project to the organisation. This is typically associated with a customer journey as the focus is on the value that customers receive and then in turn the benefits (financial, market growth) that the organisation obtains. What is not evident from the interviews is the continuous re-assessment of the business case. The benefits were evaluated at regular intervals and reprioritisation of projects and value streams did occur, but no evidence was found whether an updated business case informed these decisions or whether this was done as part of the PI. This phenomenon is in line with theory stating that the business case is seldom revisited (Einhorn et al., 2019). The same challenge is experienced with BRM in traditional project management (Marnewick, 2016). More research is needed to understand why the business case is not revisited in practice and why is there a misalignment between theory and practice.

The PI plays an important role during the agile BRM process and the results from the interviews differ slightly from the theoretical BRM process as depicted in Table 3. The PI is used for the identification, planning and execution of benefits within a scaled agile environment. BRM is not explicitly mentioned in a PI. In practice, it implies that the BRM process needs to be explicitly mentioned and managed. From a theoretical perspective, the process in Table 3 covers most of the processes but the PI needs to be formally part of the BRM process. Organisations are incorporating BRM into the PI sessions, thus merging BRM and scaling agile. This addresses the concern of Badewi (2016) and Holgeid and Jørgensen (2020) that BRM and project management processes should be merged.

Tracking of benefits was not found to be an activity within the agile benefits realisation process. This activity was highlighted by all the interviewees and should be incorporated into the process. Tracking of benefits is highly dependent on the metrics associated with a benefit (Zwikaël et al., 2018). Measurement needs to be in place if proper benefits tracking is to be done. To track benefits, the information regarding the benefits needs to be accurate and the source of the information needs to be credible and trustworthy. Tracking should focus on the effective use of resources, i.e. zero waste, as well as budget and cost. A challenge arises when a horizontal business unit such as IT implements a project for a vertical business unit. The ownership of the benefit belongs to both parties, but the results of the tracking might be different. Should a project be cancelled if one business unit harvests fewer benefits? What about the other business unit? These questions were not answered by the interviewees, who also grappled with the complexity of benefits tracking from various perspectives. The tracking of benefits typically takes place at a PI and based on the results, reprioritisation is done, and the value streams are adjusted. Course correction needs to take place, thus ensuring optimal benefits realisation.

The end-result of benefits tracking is the reprioritisation of projects. Reprioritisation takes place at two instances. The first instance is at the PI where projects are reprioritised based on value and effort. The second instance is where the teams decide to reprioritise the backlog during the sprints. Reprioritisation at team level has a direct impact on reprioritisation at the *Planning and Prioritisation* level. This implies that teams should be mature to make decisions that affect the entire organisation and consequence management should be in place.

Sustainment of benefits did not feature throughout the interviews. The reason for this is debatable, but no evidence was provided that this step was done and how it was done. Closing the loop to link the

anticipated benefits back to the value streams and strategies is not taking place. This again correlates with the research of Marnewick (2016). The sustainment of benefits is a concern in both traditional and agile BRM. It is hypothesised that organisations are not sustaining benefits as they are already reporting excessive profits as per their annual financial reports. The effort to sustain benefits does not warrant the resources to manage this step. This also speaks to waste and the question that shareholders should ask is how much waste is not dealt with during value creation.

The fact that the organisations derived value from their IT projects, can also be contributed to the actual application of the agile values and principles at the *Implementation* level. An analysis on whether the values and principles support agile BRM highlights that the Agile Manifesto is focused on short-term success and not on the long-term realisation of benefits. Only principle one focuses on the long-term satisfaction that customers receive from valuable software. This relates to dimension five ('User satisfaction') of the D&M IS Success Model (Petter et al., 2008). The rest of the values and principles are short-term focused and relates to the 'System quality', 'Service quality' and 'Use' dimensions of the D&M IS Success Model. As per Fig. 1, this contributes to the success of the information system and the product success level. Although the Agile Manifesto intrinsically speaks to short-term success, it ultimately provides the foundation for IS and project success. But it does not directly address BRM.

5.1. Principles

An unexpected outcome of the research is the identification of principles that need to be in place to ensure the success of BRM in a scaled agile environment. "So if you're looking for that golden thread, you're not going to get it. Ah, but what you would find is principles. And principles are applied. And it's what are those principles that are critical to an organisation for realizing benefits" [PART-1]. Given this statement and the fact that there is no golden thread regarding the BRM process and the ownership, principles were derived from the interviews. This is in response to Svejvig and Schlichter (2020) and addresses the complexity of BRM. The principles are categorised across the three levels and some principles cover all three levels as illustrated in Fig. 4.

5.1.1. Portfolio level

Six principles are identified for the portfolio level.

- Beyond budgeting:** Beyond budgeting is defined as an alternative performance management model that enhances organisational adaptability and responsiveness by incorporating changes in the business culture and entire management control system (Nguyen et al., 2018; O'Grady et al., 2017; Sandalgaard, 2012). The focus of beyond budgeting is a management model that is more empowered and adaptive and that is beyond the traditional command-and-control. "So the principle of beyond budgeting is that as an organisation you've identified a set of measures that would reflect whether the organisation is being successful or not" [CVO-1]. This principle is unique to agile BRM as budget allocation moves from a project-driven allocation to a fixed capacity allocation (Petit & Marnewick, 2021a).
- Course correction:** Rapid corrective action needs to be taken once the organisation realises that the intended benefits are not going to be realised. This rapid corrective action needs to be supported by agile and accounting processes. "If you starting to see if you started to see the uptake in customers but it's not correlating in a financial outcome, you should be able to course correct quite rapidly to say, well, why did I go wrong and how do I correct the process" [CVO-1]. The principle is applicable to both traditional and agile BRM. In the case of agile BRM, course correction happens at every PI and not at the end of the project as with a typical Waterfall project.
- Define target outcomes:** Effective BRM is highly dependent on well-defined targets (Zwikaal et al., 2018). The interviewees highlighted

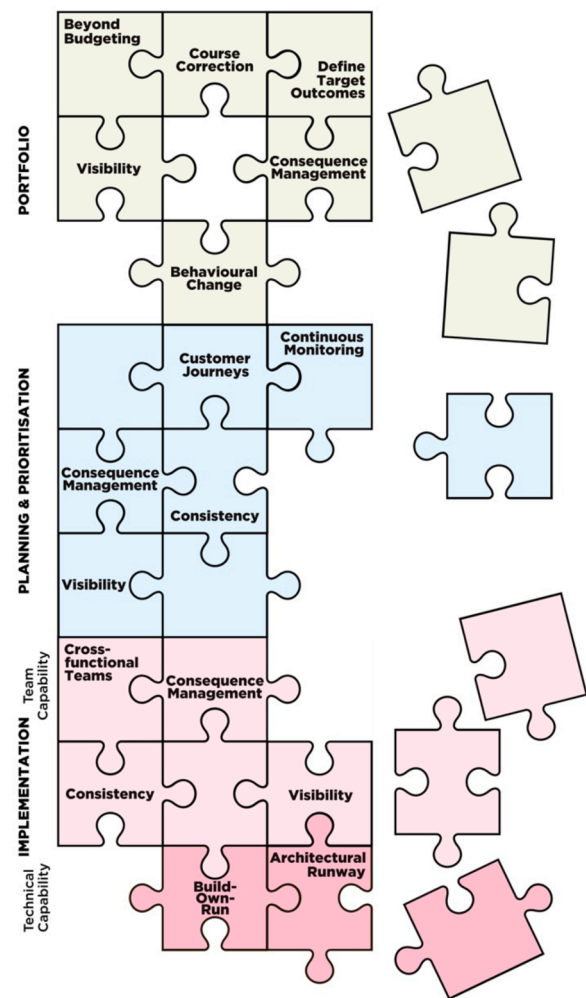


Fig. 4. Agile benefits realisation principles.

the notion that benefits tracking and prioritisation can only be done with well-defined targets. "Uh, and then looking at the metrics and targets that would be tracked around that use case once it's delivered, and then some sort of measurement depending on what it is" [TRAN-2]. There is a concern as per COACH-1: "But unfortunately it is a bit like a fox watching his own hen house because they define the metrics up front and then they also measure themselves." This principle is applicable to both traditional and agile BRM.

- Visibility:** As with traditional portfolio management, visibility focuses on showcasing the various projects and their associated value and effort (Patanakul, 2015). Effective BRM requires transparent visibility across all three levels, and this can be achieved through an enterprise visibility room (EVR). The focus of an EVR is a single view of the organisation's strategies and value streams. It visualises active strategy work and the teams engaged in delivering value. This principle also relates to VSM where the focus is on visualisation. "So through bringing visibility you've increased accountability of the people that are asking for the value to be delivered because they potentially know, or they will likely be asked to demonstrate what this is meant for the organisation, etc." [CONS-1]. This principle is specific to agile BRM as it speaks VSM is forms the basis of scaled agile.
- Behavioural change:** People need to think differently around benefits and value creation. The focus should be on adding meaningful change to the measures and metrics. "But what I have started to see is through this cadence we've actually started to see a change in behaviours with people are starting to think differently" [CVO-1]. Behavioural

change is a principle that is applicable to both traditional and agile BRM.

- 6 **Consequence management:** This principle relies on two aspects. The first is that decisions do have consequences. People are accountable for their decisions and when value is not added, the person should face the consequences. The second aspect is that decisions influence the entire organisation and BRM process. When decisions are made, all consequences need to be evaluated to determine the positive and negative impacts of the decision. Consequence management cuts across all three levels as no one is exempt from the consequences of their decisions. “If you said you’re going to do the X, why didn’t you deliver X? What have you done to course, correct? Well, if you haven’t cost corrected, well there needs to be consequences for it, right?” [CVO-1]. This principle is applicable to both traditional and agile BRM and should be informed by who the benefits owner is [Zwikaël et al. \(2019\)](#).

There is a relationship between these principles, but the strength and correlations are not known. This will be the focus of future research.

5.1.2. Planning and prioritisation level

At this level, five principles are identified.

- 1 **Customer journeys:** A customer journey focuses on customers’ holistic experience with the organisation. It involves a visual depiction of the sequence of events through which customers interact with the organisation ([Rosenbaum et al., 2017](#)). Benefits need to be mapped to customer journeys to maximise the value of the investments, but also to create value for the customer. This principle also ties closely with VSM where the focus is the customer: “... but then making sure that the customer journey is less painful and those would be there’s some qualitative and quantitative benefits in that so flexibility around revenue.” [PART-1]. TRAN-1 cautions that “knowing who your customer is and what the value you give to your customer is a very difficult concept”. This principle is unique to agile BRM.
- 2 **Continuous monitoring:** Value and effort associated with projects need to be continuously assessed. This is typically done at a PI and the purpose is to determine whether the projects are still on track to deliver the intended benefits. If this is not the case, then reprioritisation needs to take place. This principle applies to both traditional and Agile BRM but in the case of Agile BRM, continuous monitoring occurs more frequently through the sprints and PI’s.
- 3 **Consistency:** [Bass \(2016\)](#) highlights the importance of ceremonies within a scaled agile environment. Ceremonies need to be performed in a consistent way and all teams must perform all the ceremonies. Consistency in BRM-related processes such as PIs provides consistent results. CVO-1 highlighted the importance of consistency: “Consistency you’ve got to have those ceremonies, right? So once a quarter, what is the process, what’s happened? What’s being delivered?” Although, this principle is also applicable to both traditional and agile BRM, it is more applicable to agile BRM because of ceremonies like daily stand-ups and PI’s.

The principles *Consequence management* and *Visibility* are also included at this level.

5.1.3. Implementation level

This level is divided into two sub-levels, i.e. team capability (four principles) and technical capability (three principles). At team capability level, *Cross-functional teams* is a new principle. *Consequence management*, *Consistency* and *Visibility* are principles that are repeated.

- **Cross-functional teams:** These teams are responsible for the actual delivery of the projects. They should be empowered to make their own decisions ([Salameh & Bass, 2020](#)). These decisions include ways of working, prioritisation of work and how they align with the

organisational strategies. As per ([Manos, 2006](#)), cross-functional teams are at the heart of VSM and should also be a core principle of BRM. “So you would have cross functional teams that would be operating within the business uhm, affecting that change, so that is quite a big operating model change” [PART-1]. This principle is specific to agile BRM as business and IT are incorporated into cross-functional teams that forms the foundation of agile projects.

Again, there is a level of relationship between the cross-functional team, the consequences associated with their decisions, the consistency of the team’s ceremonies and the visibility of their work through Kanban boards.

Technical capability consists of the following principles:

- 1 **Build-Own-Run:** Teams need to take ownership of their work. This philosophy of ownership introduces agility into the production environment and teams understand the consequences (positive or negative) if they deploy products that are not fully functional; it detracts from the value of the customer journey. “But we also are teams on our doing you build it you run it you own it” [IT-1]. This was emphasised by PART-1 stating that “is quite a big operating model change because you are now building and running at the same time”. This is an agile BRM specific principle and is based on the notion of DevOps ([Teixeira et al., 2020](#)) that is different from Waterfall projects where there is a distinct difference between the operational and development teams.
- 2 **Architectural runway:** This principle focuses on existing code, components and technical infrastructure that are required to support the continuous flow of value ([Leffingwell, 2018](#)). The architectural runway provides the technical foundation to develop business initiatives and to implement new capabilities. This relates to item 4 (technical debt) which influences the flow of value to customers ([Cunningham, 1992](#)). “So the benefit from an engineering point of view should be simplified IT architecture, quicker times to markets” [CVO-1]. This is an agile BRM specific principle.

The principles form the foundation of agile BRM and without the principles, agile BRM is not performed optimally and thus contributes indirectly to the success of the agile project.

Project success is determined at various levels, and this is also the case within a scaled agile environment. Benefits and value associated with projects are typically determined after project delivery as per [Table 1](#). Value is continuously delivered in a scaled agile environment through the continuous delivery of a product at implementation level. Value is also continuously delivered and re-evaluated during PI sessions. This is slightly different as success, based on benefits and value, is continuously determined and course corrections can be made at regular intervals. Value is thus optimised while waste and effort are minimised. Of concern is still the lack of a proper BRM process that might negatively influence whether all benefits are realised and sustained.

5.2. Key findings

The following key findings can be made from the analysis of the results:

- 1 The agile BRM process is in principle the same as the traditional BRM process. The difference is that agile BRM is more adaptive due to the nature of agile and scaled agile.
- 2 The value case is still a problem and this is supported by [Einhorn et al. \(2022\)](#) stating that the business case is still not used throughout the project lifecycle. The results from the interviewees support this view where the value case is used for the identification of benefits but it is not used for benefits tracking and sustainment.

- 3 Benefits tracking and sustainment is still a problem even within agile BRM. Traditional BRM also do not track and sustain the benefits (Marnewick, 2017).
- 4 The identified principles contribute to a better agile BRM process. The relationship between the principles and agile BRM still needs to be investigated in future research.
- 5 Agile BRM contributes to project success. Projects are perceived as successful based on the performance of the companies within this study. What the causal relationship is, is still not clear and forms the basis for future research.

The theoretical process (Fig. 3) as well as the comparison between traditional and agile BRM (Table 3) indicates that there is no difference between the two processes. Benefits will be harvested from agile projects whether a BRM process is followed or not. This is attributed to the nature of the agile approach (Khoza & Marnewick, 2020; The Standish Group, 2018). To improve project success, benefits realisation must be incorporated into the PI. This answers the second research question.

The review of the literature clearly indicates that BR is a measurement of IT project success (Bannerman & Thorogood, 2012; DeLone & McLean, 2003; Petter et al., 2008). The results do not explicitly support this dependency as no evidence was found that BRM improved the success of the projects. This partially answers the first research question – literature suggests one thing, but the results suggest something different Table 4.

This research could not satisfactorily answer the research problem i.e. whether BRM in a scaled agile environment ensures IT project success. This is because BRM is not done as a formal process. It can therefore not be concluded that BRM contributes to IT project success. The results highlight that principles are more likely to influence benefits realisation instead of a formal agile BRM process. This is in line with current project management standards that are principle-based instead of process-based.

6. Conclusion

This article focused on BRM within a scaled agile environment. BRM in a scaled agile environment follows the same process as within a non-agile environment. The difference is that BRM in an agile environment is a continuous process where value is continuously measured. Due to this continuous delivery cycle at the *Implementation* level, benefits are inherently realised. But, agile BRM is done just as badly as BRM in traditional projects. To improve agile BRM, certain principles were derived. It is believed that these principles contribute to the successful management of benefits in a scaled agile environment.

6.1. Contribution of the study

Benefits realisation is key to the success of an IT project. This is established through the success levels of Bannerman and Thorogood (2012) as well as the success dimensions of Petter et al. (2008). The more benefits are realised, the more successful the IT project will be. The results support the theory that BRM contributes to project success.

With regards to the research question, it is established that agile BRM follows the same process as traditional BRM. The challenge is still that benefits transition and sustainment is not done as with traditional BRM. Two aspects were identified that contribute to the theory on agile BRM. Firstly, the role of the PI is fundamental in agile BRM and secondly, the identified principles should contribute to a more robust and consistent agile BRM process.

In essence, the only difference between agile BRM and traditional BRM is that in the case of agile BRM, deviations are detected earlier due to the adaptive and iterative approach. This allows then for course correction to maximise the benefits.

6.2. Limitations of the study

The major limitation of the study is the number of interviewees. Ideally, more people should have been interviewed from a specific organisation as well as from different organisations. This should remove potential bias introduced by the interviewees.

6.3. Avenues for future research

Future research will focus on three areas. The first is to understand why the entire BRM process is not followed as this is a continuous problem within IT projects. The second area will be more quantitative of nature to determine any causal relationships between agile BRM and IT project success since agile BRM is a dimension of IT project success. The third area will investigate the role of the principles within BRM and whether there is a causal relationship.

References

- Ali, N. B., Petersen, K., & De França, B. B. N. (2015). Evaluation of simulation-assisted value stream mapping for software product development: Two industrial cases. *Information and Software Technology*, 68, 45–61.
- Ambler, S. W., & Lines, M. (2012). *Disciplined agile delivery: A practitioner's guide to agile software delivery in the enterprise*. Indianapolis, Indiana, USA: IBM Press.
- Ashurst, C. (2011). *Benefits realization from information technology*. Palgrave Macmillan.
- Aubry, M., & Hobbs, B. (2011). A fresh look at the contribution of project management to organizational performance. *Project Management Journal*, 42, 3–16.
- Baccarini, D. (1999). The logical framework method for defining project success. *Project Management Journal*, 30, 25–32.
- Badewi, A. (2016). The impact of project management (PM) and benefits management (BM) practices on project success: Towards developing a project benefits governance framework. *International Journal of Project Management*, 34, 761–778.
- Bannerman, P. L., & Thorogood, A. (2012). Celebrating IT projects success: A multi-domain analysis. In *Proceedings of the 45th Hawaii international conference on system sciences*, 4–7 Jan. (pp. 4874–4883).
- Bass, J. M. (2016). Artefacts and agile method tailoring in large-scale offshore software development programmes. *Information and Software Technology*, 75, 1–16.
- Bennington, P., & Baccarini, D. (2004). Project benefits management in IT projects – An Australian perspective. *Project Management Journal*, 35, 20–30.
- Bradley, B. (2010). *Benefit realisation management: A practical guide to achieving benefits through change*. Surrey, England: Gower Publishing Limited.
- Camilleri, E. (2011). *Project success: Critical factors and behaviours*. London: Gower Publishing.
- Castillo-Montoya, M. (2016). Preparing for interview research: The interview protocol refinement framework. *The Qualitative Report*, 21, 811–831.
- Cha, J., Newman, M., & Winch, G. (2018). Revisiting the project management knowledge framework: Rebalancing the framework to include transformation projects. *International Journal of Managing Projects in Business*, 11, 1026–1043.
- Chih, Y. Y., & Zwikael, O. (2015). Project benefit management: A conceptual framework of target benefit formulation. *International Journal of Project Management*, 33, 352–362.
- Cooke-Davies, T. (2002). The “real” success factors on projects. *International Journal of Project Management*, 20, 185–190.
- Cunningham, W. (1992). The WyCash portfolio management system. *Addendum to the proceedings on object-oriented programming systems, languages, and applications (Addendum)*. Vancouver, British Columbia, Canada: Association for Computing Machinery.
- Cunningham, W. S. (2008). Voices from the field: Practitioner reactions to collaborative research initiatives. *Action Research*, 6, 373–390.
- Davis, K. (2017). An empirical investigation into different stakeholder groups perception of project success. *International Journal of Project Management*, 35, 604–617.
- DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, 19, 9–30.
- Dikert, K., Paasivaara, M., & Lassenius, C. (2016). Challenges and success factors for large-scale agile transformations: A systematic literature review. *Journal of Systems and Software*, 119, 87–108.
- Doherty, N. F. (2014). The role of socio-technical principles in leveraging meaningful benefits from IT investments. *Applied Ergonomics*, 45, 181–187.
- Doherty, N. F., Ashurst, C., & Peppard, J. (2012). Factors affecting the successful realisation of benefits from systems development projects: Findings from three case studies. *Journal of Information Technology*, 27, 1–16.
- Ebert, C., & Paasivaara, M. (2017). Scaling agile. *IEEE Software*, 34, 98–103.
- Einhorn, F., Marnewick, C., & Meredith, J. (2019). Achieving strategic benefits from business IT projects: The critical importance of using the business case across the entire project lifetime. *International Journal of Project Management*, 37, 989–1002.
- Einhorn, F. D., Meredith, J., & Marnewick, C. (2022). A model to guide the effective use of a business case for IT projects. *International Journal of Managing Projects in Business*, 15, 36–57.

- Fernandes, G., & O'sullivan, D. (2021). Benefits management in university-industry collaboration programs. *International Journal of Project Management*, 39, 71–84.
- Gomes, J., & Romão, M. (2016). Improving project success: A case study using benefits and project management. *Procedia Computer Science*, 100, 489–497.
- Holgeid, K. K., & Jørgensen, M. (2020). Benefits management and agile practices in software projects: How perceived benefits are impacted. In E. Proper, & J. Verelst (Eds.), *Proceedings of the IEEE 22nd conference on business informatics (CBI)*. IEEE.
- Holgeid, K. K., Jørgensen, M., Sjøberg, D. I. K., & Krogstie, J. (2021). Benefits management in software development: A systematic review of empirical studies. *IET Software*, 15, 1–24.
- Ika, L. A. (2009). Project success as a topic in project management journals. *Project Management Journal*, 40, 6–19.
- Joseph, N., Erasmus, W., & Marnewick, C. (2014). The idle state of information and communication technology project management. *Journal of African Business*, 15, 184–196.
- Joseph, N., & Marnewick, C. (2014). Structured equation modeling for determining ICT project success factors. In C. Messikomer (Ed.), *Proceedings of the PMI research and education conference 2014*. Project Management Institute.
- Jugdev, K., & Müller, R. (2005). A retrospective look at our evolving understanding of project success. *Project Management Journal*, 36, 19–31.
- Kazmi, S. S., Rafi, A., & Hassan, M. (2016). Impact of benefit realization management on two-dimensional model of project success: Evidence from Pakistani telecom industry. *International Review of Management and Marketing*, 6, 32–37.
- Khoza, L., & Marnewick, C. (2020). Waterfall and Agile information system project success rates – A South African perspective. *South African Computer Journal*, 32, 43–73.
- Khoza, L., & Marnewick, C. (2021). Challenges and success factors of scaled agile adoption – A South African perspective. *The African Journal of Information Systems*, 13, 164–185.
- Khurum, M., Petersen, K., & Gorschek, T. (2014). Extending value stream mapping through waste definition beyond customer perspective. *Journal of Software: Evolution and Process*, 26, 1074–1105.
- Kniberg, H., & Ivarsson, A. (2012). *Scaling Agile @ Spotify with tribes, squads, chapters & guilds* [Online]. Available: <http://www.agileleanhouse.com/lib/lib/People/HenrikKniberg/SpotifyScaling.pdf> [Accessed 23 April 2021 2021].
- Kwok, J. Y. C., & Ku, H. B. (2008). Making habitable space together with female Chinese immigrants to Hong Kong: An interdisciplinary participatory action research project. *Action Research*, 6, 261–283.
- Labuschagne, L., & Marnewick, C. (2009). *IT project management maturity vs. project success in South Africa*. The Prosperus Report 2008. Johannesburg: Project Management South Africa.
- Lappe, M., & Spang, K. (2014). Investments in project management are profitable: A case study-based analysis of the relationship between the costs and benefits of project management. *International Journal of Project Management*, 32, 603–612.
- Lappi, T., Karvonen, T., Lwakatare, L. E., Aaltonen, K., & Kuvaja, P. (2018). Toward an improved understanding of agile project governance: A systematic literature review. *Project Management Journal*, 49, 39–63.
- Larman, C., & Vodde, B. (2016). *Large-scale scrum: More with LeSS*. Boston, USA: Addison-Wesley Professional.
- Leffingwell, D. (2011). *Agile software requirements: Lean requirements practices for teams, programs, and the enterprise*. Boston: Addison-Wesley.
- Leffingwell, D. (2018). *SAFe 4.5 reference guide: Scaled agile framework for lean enterprises*. New York, U.S.A.: Addison-Wesley Professional.
- Lin, C., & Pervan, G. (2003). The practice of IS/IT benefits management in large Australian organizations. *Information & Management*, 41, 13–24.
- Longman, A., & Mullins, J. (2004). Project management: Key tool for implementing strategy. *Journal of Business Strategy*, 25, 54–60.
- Manos, T. (2006). Value stream mapping—an introduction. *Quality progress*. Milwaukee: American Society for Quality.
- Marnewick, C. (2012). A longitudinal analysis of ICT project success. In *Proceedings of the South African institute for computer scientists and information technologists conference* (pp. 326–334). ACM.
- Prosperus report - In Marnewick, C. (Ed.), *The African edition*. Johannesburg: Project Management South Africa.
- Marnewick, C. (2016). Benefits of information system projects: The tale of two countries. *International Journal of Project Management*, 34, 748–760.
- Marnewick, C. (2017). The reality of adherence to best practices for information system initiatives. *International Journal of Managing Projects in Business*, 10, 167–184.
- Marnewick, C., Erasmus, W., & Joseph, N. (2017). *The symbiosis between information system project complexity and information system project success*. Cape Town, South Africa: AOSIS (Pty) Ltd.
- McLeod, L., Doolin, B., & Macdonell, S. G. (2012). A perspective-based understanding of project success. *Project Management Journal*, 43, 68–86.
- Mir, F. A., & Pinnington, A. H. (2014). Exploring the value of project management: Linking project management performance and project success. *International Journal of Project Management*, 32, 202–217.
- Neve, J. R., Godbole, K., & Neve, R. (2017). Productivity and process improvement using 'Scaled Agile' approaches: An emphasized analysis. In *Proceedings of the international conference on inventive computing and informatics (ICICI)*, 23–24 Nov. (pp. 793–798).
- Nguyen, D. H., Weigel, C., & Hiebl, M. R. W. (2018). Beyond budgeting: Review and research agenda. *Journal of Accounting & Organizational Change*, 14, 314–337.
- O'grady, W., Akroyd, C., & Scott, I. (2017). Beyond budgeting: Distinguishing modes of adaptive performance management. *Advances in Management Accounting*.
- Patanakul, P. (2015). Key attributes of effectiveness in managing project portfolio. *International Journal of Project Management*, 33, 1084–1097.
- Peterson, M. (2002). *Why are we doing this project?* Detroit: PriceWaterhouseCoopers.
- Petit, Y., & Marnewick, C. (2021a). Fixed capacity and beyond budgeting – A symbiotic relationship within a scaled agile environment. In *Proceedings of the EURAM Conference*.
- Petit, Y., & Marnewick, C. (2021b). Strategic alignment of information technology initiatives in a scaled agile environment. *The Journal of Modern Project Management*, 8, 6–19.
- Petter, S., Delone, W., & Mclean, E. (2008). Measuring information systems success: Models, dimensions, measures, and interrelationships. *European Journal of Information Systems*, 17, 236–263.
- Pinto, J. K., & Slevin, D. P. (1987). Critical factors in successful project implementation. *IEEE Transactions on Engineering Management*, EM-34, 22–27.
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® Guide)*. Newtown square, Pennsylvania. Project Management Institute.
- Rosenbaum, M. S., Otolara, M. L., & Ramírez, G. C. (2017). How to create a realistic customer journey map. *Business Horizons*, 60, 143–150.
- Salameh, A., & Bass, J. M. (2020). Heterogeneous tailoring approach using the spotify model. In *Proceedings of the evaluation and assessment in software engineering*. Association for Computing Machinery.
- Sandalgaard, N. (2012). Uncertainty and budgets: An empirical investigation. *Baltic Journal of Management*, 7, 397–415.
- Savolainen, P., Ahonen, J. J., & Richardson, I. (2012). Software development project success and failure from the supplier's perspective: A systematic literature review. *International Journal of Project Management*, 30, 458–469.
- Scaled Agile Inc. (2020). *Value streams*. Scaled Agile, Inc. [Online] Available: <https://www.scaledagileframework.com/value-streams/> [Accessed 2 April 2020].
- Serra, C. E. M., & Kunc, M. (2015). Benefits realisation management and its influence on project success and on the execution of business strategies. *International Journal of Project Management*, 33, 53–66.
- Serrador, P., & Turner, R. (2015). The relationship between project success and project efficiency. *Project Management Journal*, 46, 30–39.
- Shenhar, A. J. (2001). One size does not fit all projects: Exploring classical contingency domains. *Management Science*, 47, 394–414.
- Shenhar, A. J., Dvir, D., Levy, O., & Maltz, A. C. (2001). Project success: A multidimensional strategic concept. *Long Range Planning*, 34, 699–725.
- Shenhar, A. J., Levy, O., & Dvir, D. (1997). Mapping the dimensions of project success. *Project Management Journal*, 28, 5–13.
- Sillitti, A., & Succi, G. (2005). Requirements engineering for agile methods. In A. Aurum, & C. Wohlin (Eds.), *Engineering and managing software requirements*. Springer Berlin Heidelberg.
- Sutherland, J. (2020). *The Scrum@Scale Guide*. Scrum Inc.
- Svejvig, P., & Schlichter, B. R. (2020). The long road to benefits management: Toward an integrative management model. *Project Management Journal*, 51, 312–327.
- Teixeira, D., Pereira, R., Henriques, T., Silva, M. M. D., Faustino, J., & Silva, M. (2020). A maturity model for DevOps. *International Journal of Agile Systems and Management*, 13, 464–511.
- Terlizzi, M. A., Albertin, A. L., & De Moraes, H. R. D. O. C. (2017). IT benefits management in financial institutions: Practices and barriers. *International Journal of Project Management*, 35, 763–782.
- The Standish Group. (2013). *Chaos manifesto 2013*. The Standish Group.
- The Standish Group. (2018). *The CHAOS report: decision latency theory*. The Standish Group.
- Theobald, S., Schmitt, A., & Diebold, P. (2019). Comparing scaling agile frameworks based on underlying practices. Hoda, R., ed. In *Proceedings of the Agile processes in software engineering and extreme programming – Workshops*, 21–25 May (pp. 88–96) Montreal, Canada: Springer International Publishing.
- Todorović, M. L., Petrović, D. Č., Milić, M. M., Obradović, V. L., & Bushuyev, S. D. (2015). Project success analysis framework: A knowledge-based approach in project management. *International Journal of Project Management*, 33, 772–783.
- Ul Musawir, A., Serra, C. E. M., Zwikael, O., & Ali, I. (2017). Project governance, benefit management, and project success: Towards a framework for supporting organizational strategy implementation. *International Journal of Project Management*, 35, 1658–1672.
- Versionone Inc. (2019). *The 13th annual state of agile Report*. VersionOne.
- Ward, J., & Daniel, E. (2012). *Benefits management: how to increase the business value of your IT projects*. United Kingdom: John Wiley & Sons.
- Wateridge, J. (1998). How can IS/IT projects be measured for success? *International Journal of Project Management*, 16, 59–63.
- Womack, J. P. (2006). Value stream mapping. *Manufacturing Engineering*, 136, 145–156.
- Yin, R. K. (2017). *Case study research: Design and methods*. Los Angeles (CA): Sage.
- Young, R., & Grant, J. (2015). Is strategy implemented by projects? Disturbing evidence in the State of NSW. *International Journal of Project Management*, 33, 15–28.
- Zwikael, O., Chih, Y. Y., & Meredith, J. R. (2018). Project benefit management: Setting effective target benefits. *International Journal of Project Management*, 36, 650–658.
- Zwikael, O., Meredith, J. R., & Smyrk, J. (2019). The responsibilities of the project owner in benefits realization. *International Journal of Operations & Production Management*, 39, 503–524.